

Final exam review

- 1, The foundations: logic and proofs
 - 1.1 propositional logic
 - 1.3 propositional equivalences
 - 1.4 predicates and quantifiers
 - 1.6 Rules of Inference
- 2, sets, functions, sequences, sums, matrices
 - 2.1 sets
 - 2.2 set operations
 - 2.3 Functions
 - 2.5 cardinality of sets
 - 2.6 matrices

- 3, algorithms
 - 3.1 the growth of functions
 - 3.3 complexity of algorithm
- 4, number theory and cryptography
 - 4.1 divisibility and modular arithmetic
 - 4.2 integer representations
 - 4.3 primes and greatest common divisors (Euclidean algorithm)
- 5, Induction and recursion
 - 5.1 mathematical induction
- 6, counting
 - 6.1 the basics of counting
 - 6.2 the pigeonhole principle
 - 6.3 permutations and combinations

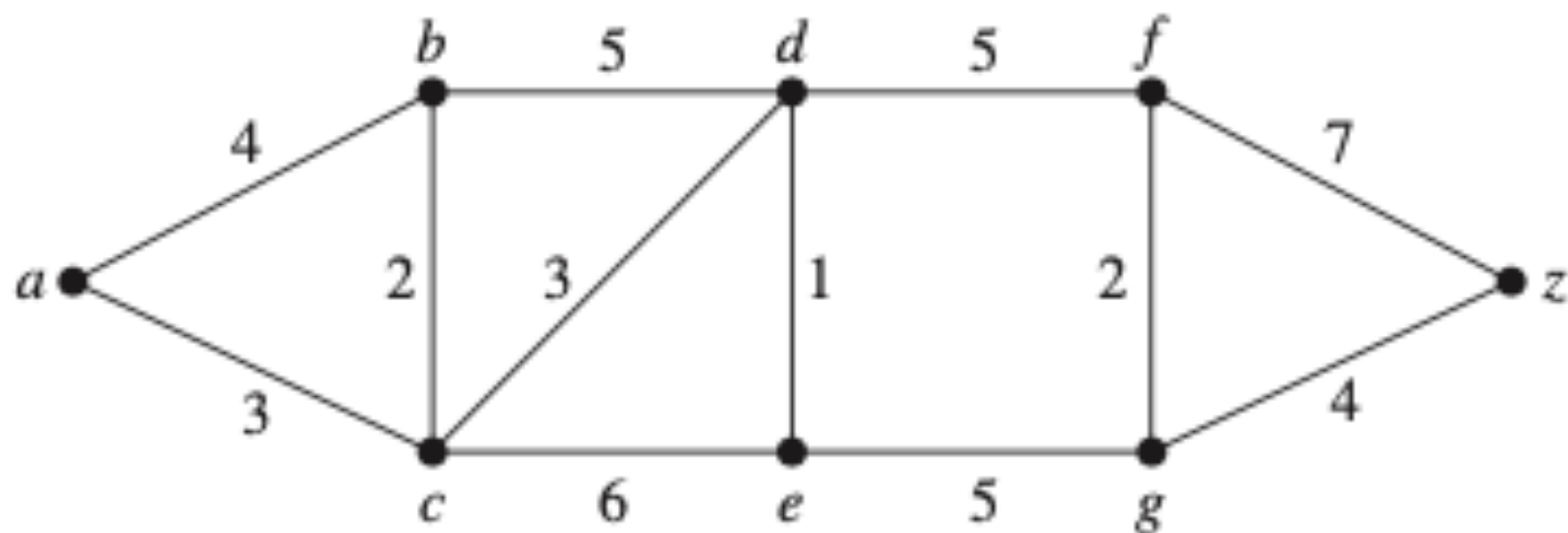
- 7. Discrete Probability
- 7.1 An introduction to discrete probability
- 7.2 Probability Theory
- 7.4 Expected Value and Variance
- 9, relations
- 9.1 relations and their properties
- 9.3 representing relations
- 10, graphs
- 10.1 graphs and graph models
- 10.2 graph terminology and special types of graphs
- 10.3 representing graphs
- 10.4 connectivity
- 10.6 shortest path problems

- 11, trees
- 11.1 Introduction to trees
- 11.2 Applications of trees (only Huffman coding)
- 11.3 Tree traversal
- 11.5 Minimum spanning trees

10.6

In Exercises 2–4 find the length of a shortest path between a and z in the given weighted graph.

3.



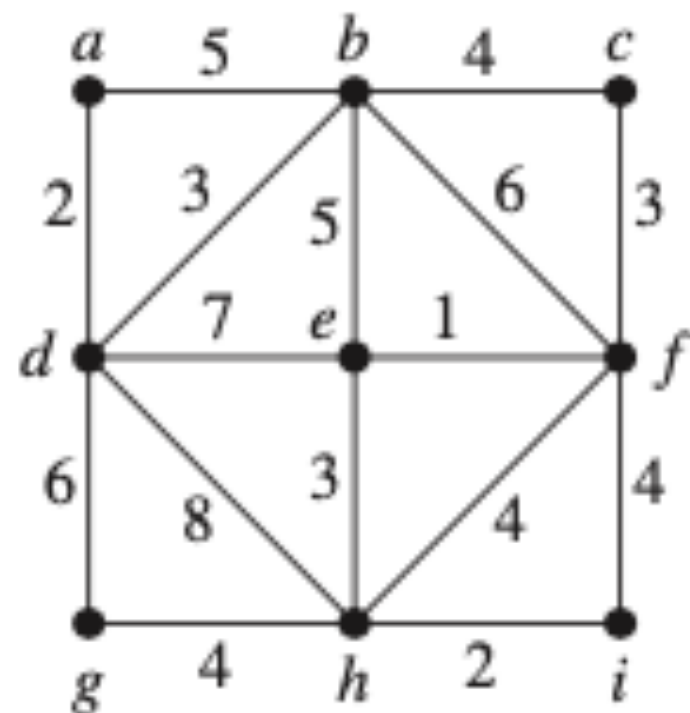
Minimum Spanning Tree

最小支撐樹

11.5

In Exercises 2–4 use Prim’s algorithm to find a minimum spanning tree for the given weighted graph.

3.



7. Use Kruskal’s algorithm to find a minimum spanning tree for the weighted graph in Exercise 3.

10.3

9. Represent each of these graphs with an adjacency matrix.

d) C_4

e) W_4

9.3

1. Represent each of these relations on $\{1, 2, 3\}$ with a matrix (with the elements of this set listed in increasing order).
 - a) $\{(1, 1), (1, 2), (1, 3)\}$
 - b) $\{(1, 2), (2, 1), (2, 2), (3, 3)\}$
 - c) $\{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$

14. Let R_1 and R_2 be relations on a set A represented by the matrices

$$\mathbf{M}_{R_1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_{R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Find the matrices that represent

- a)** $R_1 \cup R_2$. **b)** $R_1 \cap R_2$. **c)** $R_2 \circ R_1$.

9.1

6. Determine whether the relation R on the set of all real numbers is reflexive, symmetric, antisymmetric, and/or transitive, where $(x, y) \in R$ if and only if

a) $x + y = 0$.

b) $x = \pm y$.

c) $x - y$ is a rational number.

d) $x = 2y$.

e) $xy \geq 0$.

f) $xy = 0$.

g) $x = 1$.

h) $x = 1$ or $y = 1$.

7.4

- 3. What is the expected number of times a 6 appears when a fair die is rolled 10 times?
- 28. What is the variance of the number of times a 6 appears when a fair die is rolled 10 times?

8. What is the expected sum of the numbers that appear when three fair dice are rolled?

6.3

- 19.** A coin is flipped 10 times where each flip comes up either heads or tails. How many possible outcomes
- a)** are there in total?
 - b)** contain exactly two heads?
 - c)** contain at most three tails?
 - d)** contain the same number of heads and tails?

- 25.** How many ways are there for four men and five women to stand in a line so that
- a)** all men stand together?
 - b)** all women stand together?

6.2

- 17.** How many numbers must be selected from the set $\{1, 2, 3, 4, 5, 6\}$ to guarantee that at least one pair of these numbers add up to 7?

3.3

3. Give a big- O estimate for the number of operations, where an operation is a comparison or a multiplication, used in this segment of an algorithm (ignoring comparisons used to test the conditions in the **for** loops, where $a_1, a_2, \dots, a_n$ are positive real numbers).

```
 $m := 0$   
for  $i := 1$  to  $n$   
  for  $j := i + 1$  to  $n$   
     $m := \max(a_i a_j, m)$ 
```

4. Give a big- O estimate for the number of operations, where an operation is an addition or a multiplication, used in this segment of an algorithm (ignoring comparisons used to test the conditions in the **while** loop).

```
 $i := 1$   
 $t := 0$   
while  $i \leq n$   
   $t := t + i$   
   $i := 2i$ 
```